Yash Honda Quote Generation System

Technical Specification

1) Purpose

Build a secure, multi-user quote generation platform that replaces manual spreadsheet workflows and provides fast, accurate, and consistent quotations for dealership operations.

2) Current Operational Baseline

The current quotation process relies heavily on manual spreadsheet input, resulting in:

- High manual effort and slower turnaround
- Greater risk of data entry inconsistencies
- Limited real-time visibility across team members

The proposed solution addresses these gaps through centralized data management, automated pricing logic, and shared operational visibility.

3) Scope of Solution

The solution provides:

- Web-based quote generation for desktop and mobile usage
- Centralized SQL-backed quote storage
- Role-based access for sales and management users
- Quote output suitable for print/PDF
- Concurrent multi-user operation with shared visibility
- Non-production and production environments for controlled delivery

4) Core Operational Capabilities

- Capture customer and vehicle quotation inputs
- Apply pricing and business rules through backend quotation logic
- Generate structured quote outputs

- Persist quote records and activity history in SQL
- Display active daily quotes in a shared operational workspace
- Categorize quotes by point of sale using color indicators
- Support sequential visibility of concurrent quote activity
- Provide dashboard-ready reporting summaries

5) High-Level Architecture

- **Client Layer:** Desktop and mobile users
- **Web Layer:** YashHonda.com web application
- **Security Layer:** HTTPS/API security controls
- **Processing Layer:** AWS-hosted Python quotation engine
- **Data Layer:** SQL database
- **Output Layer:** Printable/PDF quotation output

6) Environment Strategy

- **Non-Production:** Development, integration, testing, and validation
- **Production:** Live dealership usage

Controlled promotion from non-production to production ensures service continuity while enabling iterative improvements.

7) API and Integration Model

- Versioned backend endpoints using `/api/v1`

  Shared API contract to align frontend and backend implementation across teams
- Standardized response and error formats
- Support for create, calculate, retrieve, open-quote view, and print/export workflows

8) Data and Storage Model

The SQL schema supports:

- Users and role assignments
- Customer records
- Vehicle/pricing reference data
- Quote headers and quote line items
- Operational status tracking and audit logs

Design supports concurrent usage, traceability, and reporting needs.

9) Security and Access

- HTTPS-only communication
- Authenticated access with role-based authorization
- Server-side validation and controlled data operations
- Auditability for operational actions

10) Non-Functional Targets

- Workflow objective: quotation process completed within the defined business time target
- Reliability: consistent quote generation and persistence
- Scalability: supports concurrent users across daily operations
- Usability: reduced typing through structured selections
- Maintainability: clear environment separation and release control

11) Implementation and Delivery Approach

- Finalize API contract and data model first
- Parallel frontend/backend build against shared contract
- Validate in non-production before production release
- Conduct team training before operational rollout

12) Expected Outcome

This design establishes a clear, scalable, and secure foundation to replace manual quotation workflows with a modern shared platform that improves operational consistency and supports future growth.

Cloud hosting server-AWS services, SQL server for database

APIS-OPENAPI

Coding language- Python for backend, front end html, css

Building space-Visual Studios (Will use github for private repository to push edits through)

•